

Homework 1:

Introduction to R

due September 14, 2026 at 10:20 (upload to Pitch2Peer portal on Brightspace --> Week 1)

This assignment involves (a) downloading and installing R, (b) going through my example R script on Brightspace called AdvStatsR_Week_1_ClassScript.R, (c) watching two instructional videos and replicating the analyses shown in the videos, and (d) importing a datafile and applying some basic commands to inspect a variable numerically and visually. While the dataframe used in the instructional videos (part c) can be found on the internet (<http://www.dangoldstein.com/flash/Rtutorial2/pedometer.csv>), the datafile that you will need to import is named parenting.sav, which can be found on Brightspace in the same folder as these instructions.

PART A: Download and install R

Of course, the first thing to do is to download and install R (if you have not yet done so). This can be done at the following website: <https://www.r-project.org/>. Click on “download R”, and you will need to select a CRAN mirror (for example the one located in the Netherlands, but it should not make a difference which one you choose). Follow the remaining instructions and you should be able to open the R statistical program at the end. If you want to, you can then also install RStudio.

If you need help to install R, you can get help via this video (if you don't need help, you do not need to watch these videos):

<http://www.youtube.com/watch?v=W101Mof0Exc&index=7&list=PLycnP7USbo1V3jlyjAzWUB201cLxPq4NP>

If you want to, you can also download and install RStudio (many students prefer to use RStudio over using R), which is a user interface for R. RStudio can be downloaded at the following website:

<https://www.rstudio.com/products/rstudio/download/>. Be sure to download one of the free versions of RStudio (e.g., RStudio Desktop). Once you have installed RStudio, you should be able to open this program, which directly accesses R (i.e., you can just start RStudio; you don't need to also start R).

The following video shows you how to install both R and RStudio:

<http://www.youtube.com/watch?v=WJDrYUqNrHg>

PART B: Recap of R commands learned in class (this will come in handy for the later parts of the homework)

Please go through my example R script from the class. I uploaded it to Brightspace under week 1; it's called something like AdvStatsR_Week_1_ClassScript.R

Do all the things I did in the example script; ideally, you don't just copy/paste my code but create your own script that is not exactly identical to what I did, but is similar (e.g., create your own vector using different/more/... numbers etc). That way you learn most.

PART C: Watch and learn!

Watch these two video by Dan Goldstein. The first video shows some basic stuff in R, the second video shows more basics plus a preview of more advanced stuff (such as linear regression analysis), which we will cover only later in the course. Watch both videos fully; and try to replicate the things that he does in the videos. Some of the important things for you to know how to do from these videos are: (1) importing an external data file, (2) using the \$ to specify a variable in a specific dataframe (as in: dataframe\$variable), (3) inspecting the structure of dataframes, (4) creating relatively simple plots (advanced ones: later classes), (5) accessing help files, (6) setting the working directory.

Video 1: <https://www.youtube.com/watch?v=ZmtkqarVTDc>

Video 2: <https://www.youtube.com/watch?v=GcmorHXygoI>

After watching these videos and working through the commands there, we are going to work with the pedometer file and compute a few things (as a reminder, you can download the file here:

<http://www.dangoldstein.com/flash/Rtutorial2/pedometer.csv>). As proof that you did these things, we are going to ask you below for some of the R code that you produce and some of the numbers that you computed (you can simply copy/paste these things from your R script into this word document).

1. **Provide the R code that (a) sets the working directory and (b) loads the 'pedometer.csv' file.**
(Please remember that setting the working directory works slightly differently for Macs vs PCs; if you can't remember, you can also have a look at my slides from the class. Also, in case the data frame seems to look wrong after importing it, one reason might be what I mentioned in the class about Dutch operating systems. Thus, if your imported file looks a bit odd, try using `read.csv2` instead of `read.csv`, and try out to specify different separators with the argument `sep =` Have a look at the class slides again, if you cannot remember the details.)

```
getwd()
setwd("/Users/heyanyan/Desktop/Master/2026-2027/2026_FA_B1/26F_R/Week 1")

df<- read.csv("pedometer.csv")
df
```

2. **Provide the R code that prints (a) the entire dataframe, (b) the first six rows (head), and (c) the last six rows (tail) of the dataframe.**

```
print(df)

head(df)

tail(df)
```

3. **What is the (a) mean and (b) standard deviation of the variable “Steps”?** Please copy/paste both the R code to compute these numbers as well as the actual mean and SD.

```
> mean(df$Steps)
[1] 9652.193

> sd(df$Steps)
[1] 3598.471
```

4. **Provide the R code that produces (a) a boxplot and (b) a histogram of the number of steps (“Steps”)**

```
boxplot(df$Steps)

hist(df$Steps)
```

5. **Provide the R code that produces a graph that indicates the number of data points for each day of the week (“Day”).** Note: This requires that “Day” is a factor; R might automatically turn Day into a factor when loading the data frame; if it did not do that automatically, you need add a new variable to the dataframe in which Day is a factor (hint: the relevant command to turn an existing variable into a factor variable is called `as.factor()`; it's best to add this new factor variable under a new name to the data frame, e.g., using `f_Day` as the name of this newly added variable). And another hint: The command to create the figure is `plot()`.

```
str(df) # chr
df$f_Day <- as.factor(df$Day)
str(df)

plot(df$f_Day)
```

6. **Provide the R code that produces a density plot of the number of steps (“Steps”).** Note: This requires the installation of the `lattice` package (please also include the R code for installing this).

```
install.packages("lattice")
library(lattice)
densityplot(df$Steps)
```

PART D: Apply what you have learned!

Most of the time you will want to analyze data sets that you collected yourself. Since we do not have time to collect data ourselves in the course, we will provide data sets for you. One dataframe that you will become familiar with in this course is called `parenting.sav`, which includes self-reports from 369 adolescents of the following variables:

PARTNO	participant number
Gender	participant sex (male = 0, female = 1)
Response	parental responsiveness (low = 0, high = 1)
Demand	parental demandingness (low = 0, high = 1)
Esteem	self-esteem (range from low = 5 to high = 30)
Control	self-control (range from low = 5 to high = 10)
Psycho	psychopathic traits (ranges from low = 10 to high = 60)
SES	socioeconomic level (ranges from low = 0 to high = 100)
Problems	internalizing and externalizing behaviors (ranges from low = 0 to high = 25)
Neurotic	neuroticism (ranges from low = 0 to high = 25)

7. Provide the R code that will import the parenting datatfile. Note: This requires the installation of the **foreign package** (please also include the R code for installing this).

```
install.packages("foreign")
library(foreign)
df2 <- read.spss("parenting.sav", to.data.frame = TRUE)
df2
```

8. Provide the R commands that print (a) the first six rows and (b) the last six rows of the dataframe.

```
head(df2)
tail(df2)
```

9. Provide the R code that produces (a) a histogram and (b) a density plot for the variable “Psycho” as well as the R code that prints the (c) mean and (d) standard deviation of the variable “Psycho.”

```
hist(df2$Psycho)

densityplot(df2$Psycho)

mean(df2$Psycho)

sd(df2$Psycho)
```

10. Create a multiple-choice exam question with 4 answer options (1 of them the correct answer, please indicate which one) that you think is a good way to test the content of the first week.

For explanation: Our plan is to ask each week that you create such a question. We hope and assume that there will be some very good questions among them and our plan is to use at least some of them (perhaps somewhat adjusted if necessary) in the actual exam. We think it might be also a good idea to compile them all and share them with everybody. We have to see what the best/most efficient way to do this would be (perhaps on Brightspace, not sure yet).

After importing a dataset into R using `read.csv()`, which of the following is the most recommended first step to verify that the data was imported correctly?

A) Run `summary(MyData)` to check summary statistics for all variables.

B) Use `head(MyData)` and `tail(MyData)` to inspect the first and last few rows.

C) Plot all variables using `plot(MyData)` to visually check for errors.

D) Immediately save the data with `write.csv()` to preserve the imported version.

Bonus assignment: Provide the R code to save the parenting data frame in your working directory as a csv file (save it without the row names).

```
write.csv(df2, "parenting.csv", row.names = FALSE)
```